

**UNIVERSITY OF THE FREE STATE
UNIVERSITEIT VAN DIE VRYSTAAT**

BLOEMFONTEIN CAMPUS/KAMPUS

CSIS3744

**DEPARTMENT: COMPUTER SCIENCE AND INFORMATICS
DEPARTEMENT: REKENAARWETENSKAP EN INFORMATIKA**

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**EXAMINATION/EKSAMEN:
Main End-of-year/Hoofjaareind 2018
PAPER/VRAESTEL 1**

**ASSESSOR(S)/
ASSESSOR(E):** 1. Mnr. W.S.J. Marais

**MODERATOR(S)/
MODERATOR(E):** 1. Dr. B.P. Haskins

TIME/TYD: 180 min

MARKS/PUNTE: 100

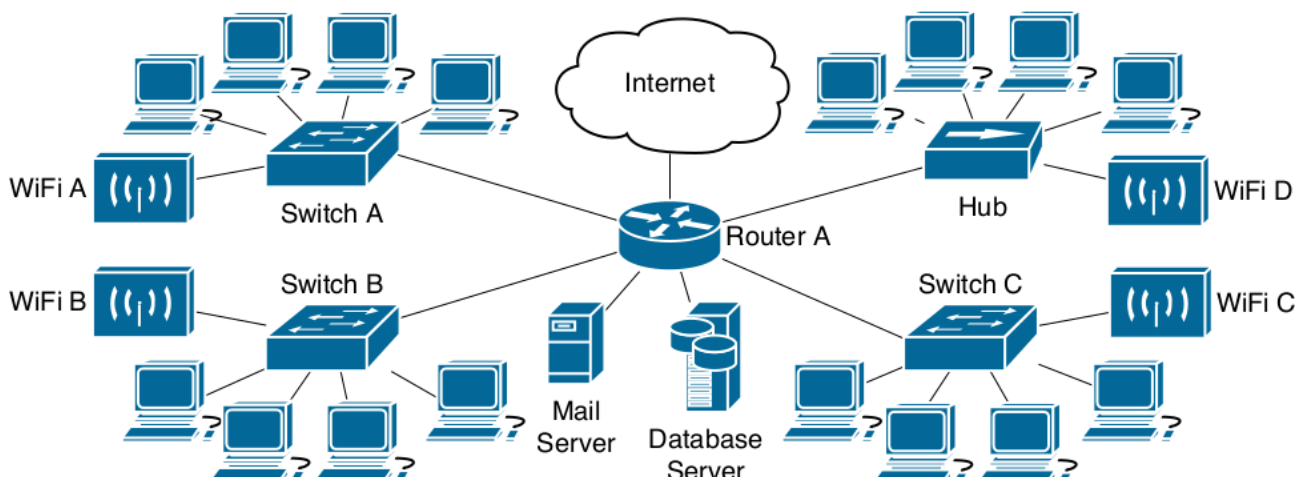
INSTRUCTIONS: Answer all the questions. Number your questions clearly and correctly. The Afrikaans versions of the questions are in italics.

INSTRUKSIES: *Beantwoord al die vrae. Nommer u antwoorde duidelik en korrek. Die Afrikaanse weergawe van die vrae is kursief gedruk.*

Question / Vraag 1 [27 Marks / Punte]

UNI-X is a small private university who implemented a client/server network on their campus. The following network diagram illustrates the layout of their network: /

UNI-X is 'n klein privaat universiteit wat 'n kliënt/bediener netwerk op hul kampus geïmplementeer het. Die volgende netwerkdiagram illustreer die uitleg van hul netwerk:



- 1.1 Indicate whether UNI-X's current network forms a LAN, MAN(CAN) or WAN? /
Dui aan of UNI-X se huidige netwerk 'n LAN, MAN(KAN) of WAN vorm? (1)
- 1.1.1 Give a motivation for your answer in 1.1. /
Gee 'n motivering vir jou antwoord in 1.1. (1)
- 1.2 Specify the physical and logical topology of UNI-X's network. /
Spesifiseer die fisiese en logiese topologie van UNI-X se netwerk. (2)
- 1.3 UNI-X attempted to implement the 1000BaseT Ethernet standard across their entire network. Unfortunately, some segments on the network could only handle the 100BaseT Ethernet standard. /
UNI-X het probeer om die 1000BaseT Ethernet-standaard oor hul hele netwerk te implementeer. Ongelukkig kon sommige segmente op die netwerk slegs die 100BaseT Ethernet-standaard hanteer.
- 1.3.1 Specify which segment(s) on the network would only be able to handle the 100BaseT Ethernet standard and explain why this is the case. /
Spesifiseer watter segment(e) op die netwerk slegs die 100BaseT Ethernet-standaard sou kon hanteer en verduidelik hoekom dit die geval is. (2)
- 1.3.2 Explain the meaning of the Ethernet standard that UNI-X attempted to implement. /
Verduidelik die betekenis van die Ethernet-standaard wat UNI-X gepoog het om te implementeer. (3)
- 1.4 Explain the difference between the operation of a hub and a switch. /
Verduidelik die verskil tussen die werking van 'n spil en 'n skakelaar. (2)
- 1.5 Compare the 1000BaseT standard with the 100BaseT standard in a table with regard to theoretical throughput, segment length, physical media and pairs of UTP wires used. /
Vergelyk die 1000BaseT-standaard met die 100BaseT-standaard in 'n tabel met betrekking tot teoretiese deurset, segmentlengte, fisiese media en pare UTP-drade wat gebruik word. (4)
- 1.6 Name and discuss the access method that is implemented by the hub. /
Noem en bespreek die toegangsmetode wat deur die spil geïmplementeer word. (5)
- 1.7 Does the hub make use of simplex, half-duplex or full-duplex transmission? /
Maak die spil gebruik van simpleks, half-dupleks of vol-dupleks transmissie? (1)
- 1.8 Specify how many collision domains and how many broadcast domains the current network contains. /
Spesifiseer hoeveel botsing-domeine en hoeveel uitsaai-domeine die huidige netwerk bevat. (2)
- 1.8.1 Explain the difference between a collision and a broadcast domain. /
Verduidelik die verskil tussen 'n botsing- en 'n uitsaai-domein. (2)
- 1.8.2 Explain the difference between throughput and bandwidth. /
Verduidelik die verskil tussen deurset- en bandwydte. (2)

[27]

Question / Vraag 2 [13 Marks / Punte]

UNI-X's wireless access points are configured as follows: /
UNI-X se draadlose toegangspunte is as volg opgestel:

- WiFi A's SSID: UniXWiFi / *WiFi A se SSID: UniXWiFi*
- WiFi B's SSID: UniXWiFi / *WiFi B se SSID: UniXWiFi*
- WiFi C's SSID: UniXWiFi / *WiFi C se SSID: UniXWiFi*
- WiFi D's SSID: UniXWiFi / *WiFi D se SSID: UniXWiFi*
- Encryption standard: WEP / *Enkripsiestandaard: WEP*
- DHCP: Enabled / *DHCP: Aangeskakel*
- Wi-Fi Protected Setup: Disabled / *WiFi Beskermd Opstelling: Afgeskakel*
- Supported 802.11 standards: /
Ondersteunde 802.11 standaarde:
 - WiFi A: n/ac
 - WiFi B: n/ac
 - WiFi C: b/g/n/ac
 - WiFi D: a/g

- 2.1 Identify and explain all the relevant security risks that are contained in the access points' configuration settings and give a solution for the vulnerabilities. /
Identifiseer en verduidelik al die tersaaklike veiligheidsrisiko's vervat in die toegangspunte se konfigurasie-instellings en gee 'n oplossing vir die kwesbaarhede. (2)
- 2.2 Explain what the "hidden node problem" entails. /
Verduidelik wat die "verborge node probleem" behels. (1)
- 2.2.1 Name and discuss the protocol that is implemented by wireless access points to counter the "hidden node problem". /
Noem en bespreek die protokol wat deur draadlose toegangspunte geïmplementeer word om die "verborge node probleem" teen te werk. (4)
- 2.3 Some users are able to connect to access points WiFi C and WiFi D, but they are unable to connect to access points WiFi A and WiFi B. /
Sommige gebruikers kan 'n verbinding met toegangspunte WiFi C en WiFi D bewerkstellig, maar kan nie met toegangspunte WiFi A en WiFi B verbind nie.
- 2.3.1 Discuss the possible pitfall that is causing the connection problem by examining the access points' configuration settings. /
Bespreek moontlike slagkat wat die verbindingsprobleem veroorsaak deur die toegangspunte se konfigurasie-instellings te ondersoek. (2)
- 2.4 Discuss the technologies, which allow 802.11n/ac to achieve such high throughput rates. /
Bespreek die tegnologieë, wat 802.11n/ac toelaat om so hoë deursetkoerse bereik. (4)
- [13]

Question / Vraag 3 [10 Marks / Punte]

Currently UNI-X's ICT services manages its IT infrastructure, provides platform services and administers all the software and data backups. Top management wants to reduce ICT's responsibilities to software administration only and suggests switching to a peer-to-peer network model to achieve this. /

Tans bestuur UNI-X se IKT-dienste sy IT-infrastruktuur, bied platformdienste en administreer al die sagteware en datarugsteunings. Topbestuur wil die verantwoordelikhede van IKT beperk tot slegs sagteware-administrasie en stel voor om na 'n eweknie-netwerkmodel oor te skakel om dit te bereik.

- 3.1 Explain to top management why a peer-to-peer network model is not an effective solution for UNI-X. /

Verduidelik aan topbestuur waarom 'n eweknie-netwerkmodel nie 'n effektiewe oplossing vir UNI-X is nie.

(3)

- 3.2 Explain to top management how cloud services could provide a more effective solution by comparing the three main cloud service models and specifying which one would meet the needs of UNI-X. /

Verduidelik aan topbestuur hoe "cloud"-dienste 'n meer effektiewe oplossing kan bied deur die drie belangrikste "cloud"-diensmodelle te vergelyk en te spesifiseer watter een aan die behoeftes van UNI-X sal voldoen.

(7)

[10]

Question / Vraag 4 [25 Marks / Punte]

UNI-X has an ADSL Internet connection. Their router's internal IP address is 192.168.1.254 and its public IP address is 72.131.141.151. /

UNI-X het 'n ADSL Internet konneksie. Hul roetebepaler se interne IP-adres is 192.168.1.254 en sy publieke IP-adres is 72.131.141.151.

- 4.1 Identify the type of router that connects UNI-X's LAN to the Internet; is it an interior, exterior or border router? /

Identifiseer die tipe roetebepaler wat UNI-X se LAN aan die Internet verbind; is dit 'n interieure-, eksterieure- of grensroetebepaler?

(1)

- 4.1.1 On what level of the OSI model does a router operate? /

Op watter vlakke van die OSI-model funksioneer 'n roetebepaler?

(1)

- 4.1.2 Discuss the IS-IS link-state routing protocol. /

Bespreek die IS-IS skakel-toestand roetebepalingsprotokol.

(4)

- 4.2 What protocol is used for connecting home computers to an ISP via DSL? /

Watter protokol word gebruik vir die aansluiting van huisrekenaars aan 'n ISP via DSL?

(1)

4.3 What server is needed at the ISP's premises for users to establish an ADSL connection using the Point-to-Point Protocol (PPP)? /
Watter bediener word benodig op die ISP se perseel sodat gebruikers 'n ADSL-konneksie kan bewerkstellig met behulp van die Punt-tot-Punt-Protokol (PPP)? (1)

4.3.1 Why is Point-to-Point Protocol (PPP) a better protocol than Serial Line Internet Protocol (SLIP)? /
Waarom is Punt-tot-Punt Protokol (PPP) 'n beter protokol as Serie-Lyn Internet Protokol (SLIP)? (5)

4.4 A workstation with the IP address 192.168.1.1 sends a request for a webpage to an Internet webserver with the IP address 172.13.14.15. Thereafter the webserver responds to the workstation's request. Illustrate how NAT in combination with PAT is implemented on the router by completing the table below, specifying the source and destination socket addresses as the request and response travels across the Internet. /

'n Werkstasie met die IP-adres 192.168.1.1 stuur 'n versoek vir 'n webblad na 'n Internet webbediener met die IP-adres 172.13.14.15. Daarna reageer die webbediener op die werkstasie se versoek. Illustreer hoe NAT in kombinasie met PAT geïmplementeer word op die roetebepaler deur die onderstaande tabel te voltooi, deur die bron- en bestemming-sokadresse te spesifiseer soos wat die versoek en reaksie oor die Internet beweeg.

	Request / Response	Source socket Bron-sok	Destination socket Bestemmingsok	
4.4.1	The request travels from the workstation to the router. / <i>Die versoek beweeg van die werkstasie na die roetebepaler.</i>			(2)
4.4.2	The request travels from the router to the webserver. / <i>Die versoek beweeg van die roetebepaler na die webbediener.</i>			(2)
4.4.3	The response travels from the webserver to the router. / <i>Die reaksie beweeg van die webbediener na die roetebepaler.</i>			(2)
4.4.4	The response travels from the router to the workstation. / <i>Die reaksie beweeg van die roetebepaler na die werkstasie.</i>			(2)

4.5 Explain in detail the role that the DNS protocol played during the transmission of the request with regard to local, top level and root name servers. /
Verduidelik volledig die rol wat die DNS-protokol gespeel het tydens die oordrag van die versoek met betrekking tot plaaslike, boonste-vlak en hoof naam-bedieners. (4)
 [25]

Question / Vraag 5 [10 Marks / Punte]

UNI-X's internal network address is 192.168.1.0. They want to create two separate subnets on their network. /

UNI-X se interne netwerkadres is 192.168.1.0. Hulle wil twee afsonderlike subnetwerke op hul netwerk te skep.

- 5.1 Which connectivity device would the administrator have to reconfigure to implement the subnets? /
Watter verbindingstoestel sal die administrateur moet verstel om die subnetwerke te implementeer? (1)
- 5.2 Calculate the new subnet mask for the proposed subnets. /
Bereken die nuwe subnetmasker vir die voorgestelde subnetwerke. (4)
- 5.3 How many usable addresses are available per subnet? /
Hoeveel bruikbare adresse is beskikbaar per subnetwerk? (1)
- 5.4 Complete the table below by supplying the network address, broadcast address and usable host address range for the 2 subnets: /
Voltooi die onderstaande tabel met die netwerkadres, uitsaai-adres en bruikbare gasheer-adresreeks vir die 2 subnetwerke:

	Subnet	Network Address Netwerkadres	Broadcast Address Uitsaai-adres	Usable Host Address Range Bruikbare Gasheer-adresreeks
5.4.1	1			
5.4.2	2			

(2)

(2)

[10]

Question / Vraag 6 [15 Marks / Punte]

Study the following header and payload information of an intercepted network packet. The information has already been translated into readable text format. /

Bestudeer die volgende kopstuk- en "payload"-inligting van 'n onderskepte netwerkpakkie. Die inligting is reeds in leesbare teksformaat vertaal.

Bit 0-7		Bit 8-15			Bit 16-23			Bit 24-31		
4	20 bytes	0			40 bytes					
0						1	0	0		
60		TCP			Correct					
172.13.14.15										
192.168.1.1										
443					49153					
3043958669										
937013559										
20 bytes		0	1	0	0	1	0	3200 bytes		
Correct										

- 6.1 Which Transport layer protocol was used during the transmission of the packet? /
Watter Transportvlakprotokol is tydens die oordrag van die pakkie gebruik? (1)
- 6.2 Which Application layer protocol was used during the transmission of the packet? /
Watter Toepassingsvlakprotokol is tydens die oordrag van die pakkie gebruik? (1)
- 6.3 What is the total size of the packet? /
Wat is die totale grootte van die pakkie? (1)
- 6.4 Provide the source and destination socket addresses for the transmission. /
Verskaf die bron- en bestemming-sokadresse vir die oordrag. (4)
- 6.5 How many hops can the packet still be forwarded before it will be discarded? /
Hoeveel "hops" kan die pakkie nog steeds aangestuur word voordat dit weggegooi word? (1)
- 6.6 Explain the purpose of the "3200 bytes" value in the header. /
Verduidelik die doel van die "3200 grepe" waarde in die kopstuk. (2)
- 6.7 The packet illustrated above was the second packet of a three-way handshake transmitted between a workstation and a webserver on the Internet. /
Die pakkie wat hierbo geïllustreer word, was die tweede pakkie van 'n drie-rigtinghanddruk wat tussen 'n werkstasie en 'n webbediener op die Internet oorgedra is.
- 6.7.1 Please illustrate how the third packet of the three-way handshake will look by specifying its sequence number, acknowledgment number, source socket, destination socket and transport layer flags that will be set. /
Illustreer asseblief hoe die derde pakkie van die drie-rigtinghanddruk sal lyk deur die volgnommer, erkenningsnommer, bron-sok, bestemming-sok en transportlaagvlaggies wat ingestel sal word, te spesifiseer. (5)
- [15]

End of Paper / Einde van Vraestel